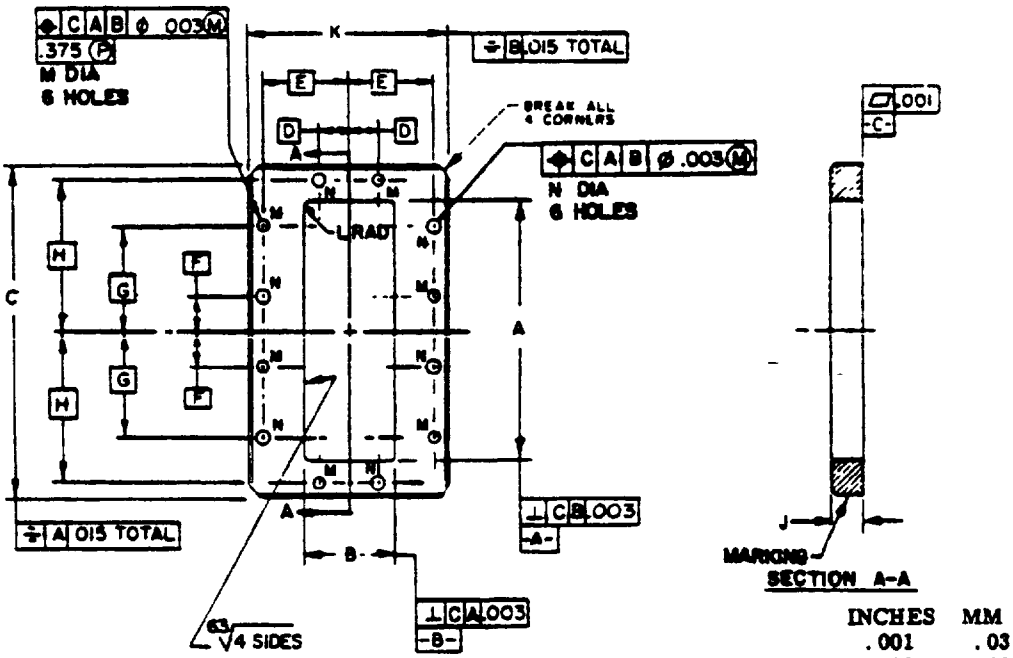
MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE, (MINIATURE, UNPRESSURIZED CONTACT)

(6 THROUGH HOLE, 6 TAPPED HOLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-3922.



INCHES	MM
.001	.03
.003	.08
.015	.38
.040	1.02
.164	4.17
.375	9.53
.75	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Roughness of mating surfaces shall not exceed 63 microinches in accordance with ANSI-B46.1 - 1962, except that flaws shall be included in the roughness height measurement.
4. For lettered dimensions, see table I.

FIGURE 1. Flange.

FSC 5985

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TABLE I. Dash numbers and dimensions.

Part No. M3922/64-			Dimensions 1/					
Dash No.	Used with Waveguide M85/1-	Used with Mating flange M3922/64-	A +.003 (.08) -.000	B +.003 (.08) -.000	C ±.015 (.38)	D BSC.	E BSC.	F BSC.
001 2/ 002 3/	043 041	001 002	3.002 (76.25)	1.502 (38.15)	3.844 (97.64)	.335 (8.51)	.995 (25.27)	.407 (10.34)

Dash No.	G BSC.	H BSC.	J ±.025 (.64)	K ±.015 (.38)	L Max	M	N +.003 (.08) -.002 (.05)
001 2/ 002 3/	1.221 (31.01)	1.745 (44.32)	.350 (8.89)	2.344 (59.54)	.010 (.25)	(No. 8) .164-32 UNC-2B	.173 (4.39)

1/ Metric equivalents are in parentheses.
2/ Copper alloy.
3/ Aluminum alloy.

FIGURE 1. Flange - Continued.

TABLE II. Items supplied with flange.

Item	M3922/64-001, 002	Quantity
Hex-head cap screw	(No. 8) .164-32 UNC-2A, .75 inch long	6
Lock washer	164 x .040 thick	6
Hex nut	(No 8) .164-32 UNC-2B	6

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material: Metallic alloy as specified in table I.

Marking: See figure 1.
Part number: M3922/64- (and a dash number from table I).

Items supplied with flanges: See table II.

AN nomenclature and engineering data: See table III.

TABLE III. Cross-reference of part numbers/AN nomenclature and engineering data.

Part No. M3922/64-	AN nomenclature	Frequency range
		GHz
001	UG-1479/U	2.60 - 3.95
002	UG-1484/U	2.60 - 3.95

Custodians:
Army-- EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, SG, EL
Navy - None
Air Force - 11, 85, 80
DSA - ES

User activities:
Army - AV
Navy - AS, OS, MC, SH, CG
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0822-4)

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a. Paragraph Number and Wording:

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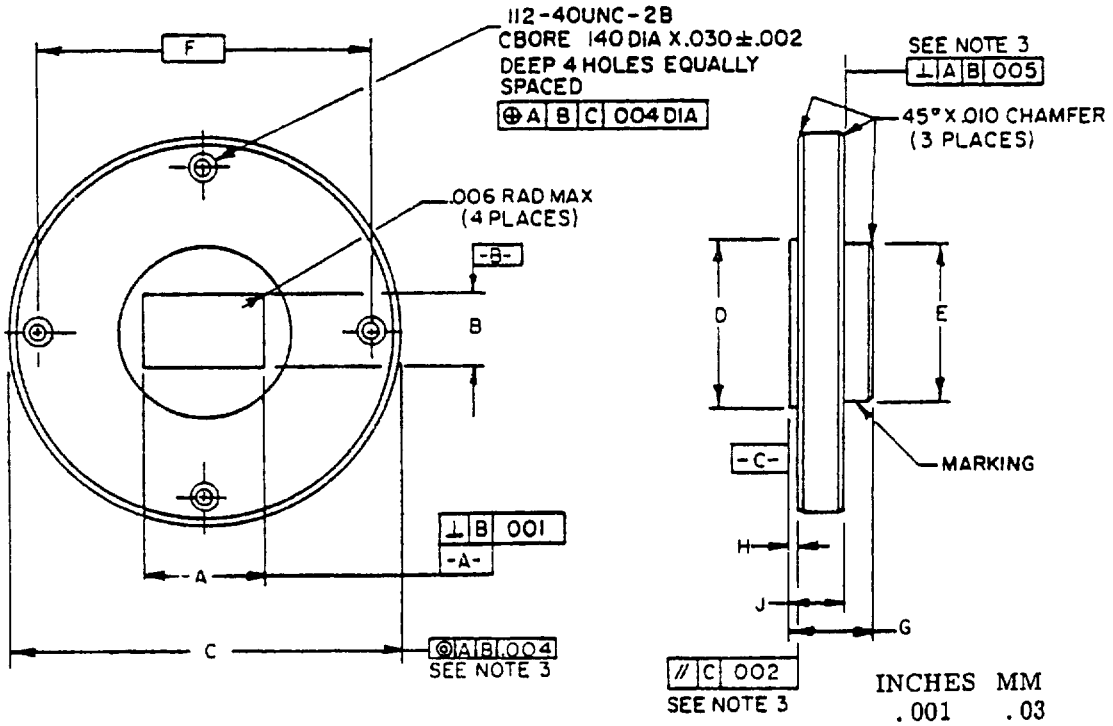
MIL-F-3922/67B
17 December 1975
SUPERSEDING
MIL-F-3922/67A
20 December 1971
MIL-F-3922/65A
20 December 1971

MILITARY SPECIFICATION SHEET

FLANGES, WAVEGUIDE (CONTACT) (ROUND, 4 HOLE) (MILLIMETER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the flanges described herein shall consist of this document and the latest issue of Specification MIL-F-3922.



INCHES	MM
.001	.03
.002	.05
.004	.10
.005	.13
.006	.15
.010	.25
.030	.76
.112	2.84
.140	3.56

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. This requirement is not applicable to dash numbers 001, 002 or 003.
4. See table I for lettered dimensions.

FIGURE 1. Flange.

FSC 5985

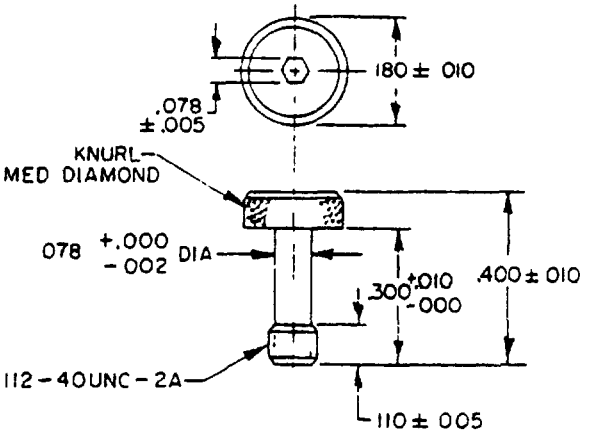
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TABLE I Flange dimensions

Part number M3922 67-			Part number M3922/67-			Dimensions 1/								
Dash no 2/	Used with Waveguide	Mating flange	Dash no 3/	Used with Waveguide	Mating flange	A +.002 (.05) -.000	B +.002 (.05) -.000	C +.000 -.002 (.05)	D ±.005 (.13)	E ±.005 (.13)	F BSC	G ±.005 (.13)	H ±.005 (.13)	J
001	M85/3-010 (RG-97/U), 011, 012 (RG-272/U)	M3922/67-001	---	---	---	.309 (7.85)	.197 (5.00)	1.125 (28.58)	.406 (10.31)	.406 (10.31)	.9375 (23.81)	.268 (6.81)	.042 (1.07)	.156 (3.96) ±.005 (.13)
002	M85/3-017 (RG-98/U), 018, 019 (RG-273/U)	M3922/67-002	---	---	---	.233 (5.92)	.159 (4.04)	.750 (19.05)	.328 (8.33)	.328 (8.33)	.5625 (14.29)	.268 (6.81)	.030 (.76)	.156 (3.96) ±.005 (.13)
003	M85/3-020 (RG-99/U), 021, 022 (RG-274/U)	M3922/67-003	---	---	---	.207 (5.26)	.146 (3.71)	.750 (19.05)	.296 (7.52)	.296 (7.52)	.5625 (14.29)	.268 (6.81)	.030 (.76)	.156 (3.96) ±.005 (.13)
004	M85/1-100, 102 (RG-53/U), 106 (RG-66/U)	M3922/67-004	011	M85/1-103 (RG-121/U), 104	M3922/67-011	.502 (12.75)	.252 (6.40)	1.125 (28.58)	.625 (15.88)	.625 (15.88)	.9375 (23.81)	.250 (6.35)	.032 (.81)	.160 (4.06) -.005(.13) -.002(.05)
005	M85/3-006 (RG-96/U), 007, 008 (RG-271/U)	M3922/67-005	012	M85/3-009	M3922/67-012	.362 (9.19)	.222 (5.64)	1.125 (28.58)	.500 (12.70)	.468 (11.89)	.9375 (23.81)	.250 (6.35)	.032 (.81)	.160 (4.06) -.005(.13) -.002(.05)
006	M85/3-010 (RG-97/U), 011, 012 (RG-272/U)	M3922/67-006	013	M85/3-013	M3922/67-013	.306 (7.77)	.194 (4.93)	1.125 (28.58)	.500 (12.70)	.468 (11.89)	.9375 (23.81)	.250 (6.35)	.032 (.81)	.160 (4.06) .005(.13) -.002(.05)
007	M85/3-014, 015, 016 (RG-358/U)	M3922/67-007	---	---	---	.270 (6.86)	.176 (4.47)	1.125 (28.58)	.500 (12.70)	.468 (11.89)	.9375 (23.81)	.250 (6.35)	.032 (.81)	.160 (4.06) -.005(.13) -.002(.05)
008	M85/3-017 (RG-98/U), 018, 019 (RG-273/U)	M3922/67-008	---	---	---	.230 (5.84)	.156 (3.96)	.750 (19.05)	.375 (9.53)	.312 (7.92)	.5625 (14.29)	.250 (6.35)	.032 (.81)	.160 (4.06) -.005(.13) -.002(.05)
009	M85/3-020 (RG-99/U), 021, 022 (RG-274/U)	M3922/67-009	---	---	---	.204 (5.18)	.143 (3.63)	.750 (19.05)	.375 (9.53)	.312 (7.92)	.5625 (14.29)	.250 (6.35)	.032 (.81)	.160 (4.06) -.005(.13) -.002(.05)
010	M85/3-023, 024, 025 (RG-359/U)	M3922/67-010	---	---	---	.182 (4.62)	.132 (3.35)	.750 (19.05)	.375 (9.53)	.312 (7.92)	.5625 (14.29)	.250 (6.35)	.032 (.81)	.160 (4.06) -.005(.13) -.002(.05)

1/ Metric equivalents are in parentheses
2/ Material shall be copper alloy
3/ Material shall be aluminum alloy

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INCHES	MM
.002	.05
.005	.13
.010	.25
.078	1.98
.110	2.79
.112	2.84
.180	4.57
.300	7.62
.400	10.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Socket-head cap screw.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material. Flange: See table I.
Cap screw: Corrosion resistant steel in accordance with QQ-S-763, type 302 or 304

Marking: See figure 1.
Part number: M3922/67-(dash number from table I).

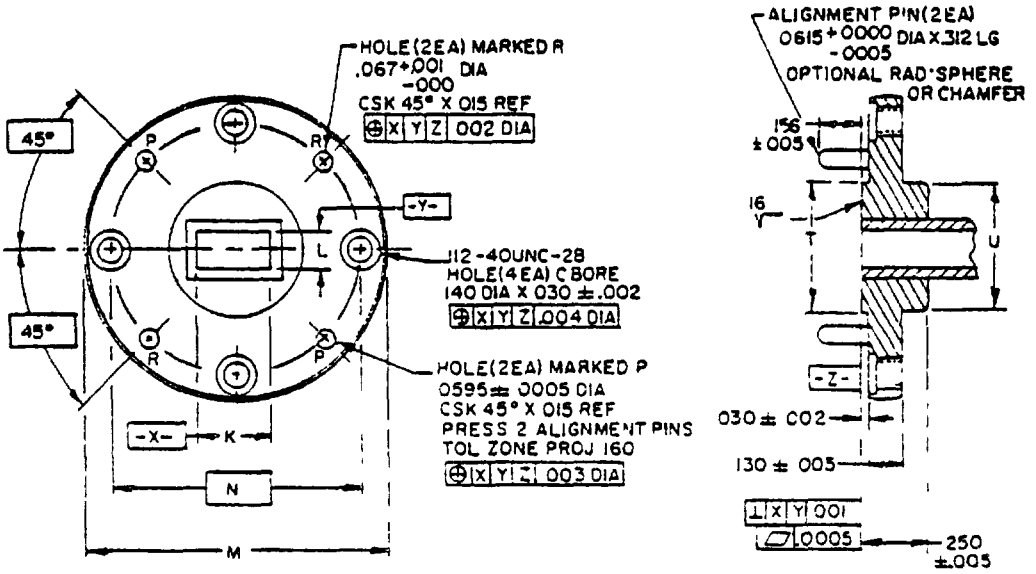
Material supplied with flange: Four socket-head cap screws as specified in figure 2.

ENGINEERING INFORMATION:

Finished flange and waveguide assembly (dash numbers 004 through 013 only). See figure 3 and table II.

Cross reference and supersession data, frequency range, and return loss data See table III.

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INCHES	MM	INCHES	MM	INCHES	MM
.0005	.01	.015	.38	.140	3.56
.001	.03	030	.76	156	3.96
.002	.05	.0595	1.51	.160	4.06
.003	.08	.0615	1 56	.250	6 35
.004	.10	067	1.70	.312	7.92
.005	.13	112	2.84		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. See table II for lettered dimensions

FIGURE 3. Finished flange and waveguide assembly.

TABLE II. Finished flange and waveguide assembly dimensions (004 thru 013 only).

Part number M3922/67-	Dimensions 1/ K ±.0015 (.04)	L ± 0015 (04)	M -.000 -.002 (05)	N BSC.	T ± 005 (13)	U ±.005 (13)
004, 011	.4200 (10.67)	.1700 (4.32)	1.125 (28.58)	9375 (23.81)	.625 (15.88)	.625 (15.88)
005, 012	2800 (7.11)	.1400 (3.56)	1.125 (28.58)	9375 (23.31)	.500 (12 70)	468 (11.89)
006, 013	.2240 (5.69)	.1120 (2.84)	1.125 (28.58)	.9375 (23.81)	.500 (12 70)	.468 (11.89)
007	.1880 (4.78)	0940 (2.39)	1.125 (28.58)	.9375 (23.81)	.500 (12.70)	.468 (11.89)
008	.1480 (3.76)	.0740 (1 88)	.750 (19.05)	.5625 (14.29)	.375 (9 53)	.312 (7.92)
009	.1220 (3.10)	0610 (1.55)	750 (19.05)	5625 (14.29)	.375 (9 53)	312 (7 92)
010	.1000 (2.54)	.0500 (1.27)	750 (19 05)	.5625 (14.29)	.375 (9.53)	312 (7.92)

1/ Metric equivalents are in parentheses

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TABLE III. Suggested frequency range, cross reference, supersession data, and return loss data.

Part number M3922/67-	AN nomenclature	Frequency range	Worst case return loss (dB)	
			Lowest frequency	Highest frequency
		GHz		
001	UG-383/U	33.0 - 50 0	-29.6	-36.3
002	UG-385/U	50 0 - 75 0	-22.4	-28.9
003	UG-387/U	60.0 - 90 0	-18 8	-25.4
004	---	18 0 - 26.5	-55 1	-61.2
005	---	26 5 - 40 0	-48.7	-54.2
006 1/	---	33.0 - 50 0	-43 7	-50 3
007 2/	---	40 0 - 60 0	-41 4	-47.4
008	---	50 0 - 75 0	-36 5	-43 0
009	---	60 0 - 90.0	-32 9	-39.5
010	---	75 0 - 110 0	-30 1	-36 1
011	---	18 0 - 26 5	-55 1	-61 2
012	---	26.5 - 40 0	-48.7	-54.2
013	---	33 0 - 50 0	-43 7	-50.3

1/ Supersedes M3922/65-001.
2/ Supersedes M3922/65-002

Custodians.
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI, MU, SG
Navy - OS
Air Force - 11, 80
DSA - ES

User activities
Army - ME
Navy - AS, MC, CG
Air Force - 14, 19

Preparing activity
Navy - EC

Agent
DSA - ES

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b. Recommended Wording.

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